

CSS 2D Transforms

- CSS 2D Transform feature allows the element to be transformed in 2-Dimensional space.
- 2D is along X-Axis and Y-Axis
- CSS provides “transform” property to defined transformations.
- CSS transforms are not supported on various browsers.
- You have to use special plug-in for different browsers.

Browser	Plugin
Chrome, Safari, Opera	-webkit
Firefox	-moz
Internet Explorer 9	-ms

Syntax:

`transform: someMethod();` `//standard`

`-webkit-transform: someMethod();`

`-moz-transform:someMethod();`

`-ms-transform:someMethod()`

-o-transform:someMethod()

Various transforms provided by CSS are:

translate()

- It is used to move the element from its current position to a new position along x and y axis.

Syntax:

transform:translate(xPixels, yPixels);

transform:translateX(pixels);

transform:translateY(pixels);

Note: Transformation happens in just one second, you can set timing for transformation by using “transition” attribute.

Ex:

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Translate</title>
```

```
<style>
  img{
    margin-left: 10px;
    transition: 2s;
  }
  img:hover{
    transform: translate(500px,200px);
    -webkit-transform: translate(500px,
200px);
    -moz-transform: translate(500px, 200px);
    -ms-transform: translate(500px, 200px);
    transition: 2s;
  }
</style>
</head>
<body>
  
</body>
```

</html>

scale()

- It is used to increase or decrease the size of element along x and y axis.
- You can control the height and width of element.

Syntax:

transform:scale(scaleX, scaleY);

transform:scaleX()

transform:scaleY()

- Scale units are defined in fractions
 - 1 = 100
 - 1.5 = 150
 - 2 = 200

Syntax:

transform:scale(2) // both x and y will be 200

transform:scale(2,1) // x will be 200 and y 100

transform:scaleX(2) // x will be 200

X = width

Y = height

Ex:

<!DOCTYPE html>

```
<html>
  <head>
    <title>Translate</title>
    <style>
      img{
        margin-left: 10px;
        transition: 2s;
      }
      img:hover{
        transform: scale(3) ;
        transition: 2s;
      }
      .container div {
        align-items: center;
        justify-content: center;
        margin: auto;
        width: 300px;
        margin-top: 200px;
      }
    </style>
  </head>
  <body>
    <div class="container">
      <div>
```

```
        
        </div>
    </div>
</body>
</html>
```

Skew()

- Skew is used to tilt the image by specified angle.

Syntax:

transform:skew(sx, sy)

transform:skewX(angle)

transform:skewY(angle)

angle clockwise: 0 to 360

angle counter clockwise: -360

Ex:

```
<!DOCTYPE html>
```

```
<html>
```

```
    <head>
```

```
        <title>Translate</title>
```

```
        <style>
```

```
            img{
```

```
        margin-left: 10px;
        transition: 2s;
        height: 300px;
        width: 300px;
    }
    img:hover{
        transform: skew(-20deg) ;
        transition: 2s;
    }
    .container div {
        align-items: center;
        justify-content: center;
        margin: auto;
        width: 300px;
        margin-top: 200px;
    }
</style>
</head>
<body>
    <div class="container">
        <div>
            
        </div>
```

```
    </div>
  </body>
</html>
```

Rotate()

- It rotates the element around its origin.
- It uses value defined in deg.
- It uses 0 to 360 deg angle
- You can use -ve value for counter clockwise.

Syntax:

```
transform:rotate('angle');
```

```
transform:rotateX()
```

```
transform:rotateY()
```

Ex:

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Translate</title>
```

```
    <style>
```

```
      img{
```

```
        margin-left: 10px;
```

```
        transition: 2s;
```

```
        height: 300px;
```

```
        width: 300px;
```

```
      }
```

```
img:hover{
    transform: rotate(-180deg) ;
    transition: 2s;
}
.container div {
    align-items: center;
    justify-content: center;
    margin: auto;
    width: 300px;
    margin-top: 200px;
}
</style>
</head>
<body>
    <div class="container">
        <div>
            
        </div>
    </div>
</body>
</html>
```

Transform Matrix

- Matrix allows to define all 2D transforms to single element.
- The transform methods
 - `translate(x,y)`
 - `translateX()`
 - `translateY()`
 - `scale(x,y)`
 - `scaleX()`
 - `scaleY()`
 - `rotate()`
 - `rotateX()`
 - `rotateY()`
 - `skew()`
 - `skewX()`
 - `skewY()`

Syntax:

`matrix(scaleX(), skewY(), skewX(), scaleY(),
translateX(), translateY())`

Ex:

`<!DOCTYPE html>`

`<html>`

```
<head>
  <title>Translate</title>
  <style>
    img{
      margin-left: 10px;
      transition: 2s;
    }
    img:hover{
      transition: 2s;
      transform: matrix(1.5,20,0.5,0,0,0);
    }

  </style>
</head>
<body>
  <div class="container">
    <div>
      
```

</div>

</div>

</body>

</html>